

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

CMOS mixed signals VLSI design [ECE_4445]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the nonidealities which affect the mixed signal design in VLSI. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Explain with diagram each block of mixed signal design. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Illustrate the effects of junction capacitance on switched capacitor design. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Design a logic to generate two non-overlapping clock signals has state sequence of 11-10-01-00. (4)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Explain the working of parasitic sensitive discrete time integrator . (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Calculate the equivalent resistance of 5pf capacitor sampled at 1 Mhz clock frequency (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Find the number of input combinations, values for 1 LSB, and the full-scale voltage generated for a 3-bit, 8-bit, and 16-bit DAC, assuming that $V_{ref} = 12\text{ V}$. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Calculate the differential nonlinearity of the 3-bit nonideal ADC whose transfer curve given in figure 3B. Assume that $V_{ref} = 5\text{ V}$ and ideal step width = 1LSB. Draw the quantization error, Q_e , in units of LSBs. (3)

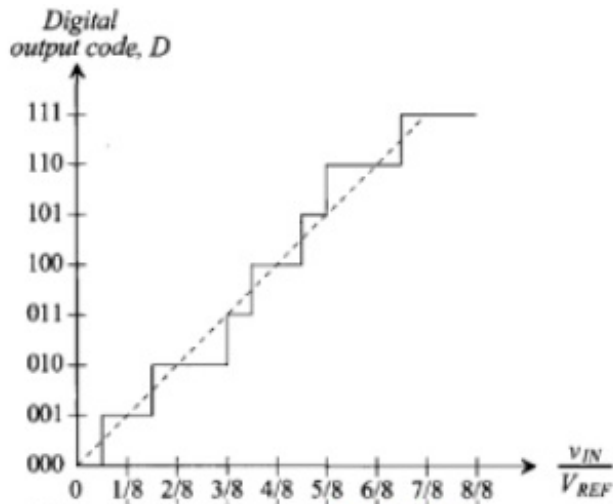


Figure 3B

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Design a 3-bit Flash converter, listing the values of the voltages at each resistor tap, and draw the transfer curve for $V_{in}=0$ to 5 V. Assume $V_{ref} = 5$ V. Construct a table listing the values of the thermometer code and the output of the decoder for $V_{in}=1.9$ V, 2.5V, and 4.1V. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Design a 3-bit charge-scaling DAC and find the value of the output voltage for D2D1D0= 011 and 110. Assume that $V_{ref}=5$ V and $C=0.5$ pF. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Explain the working of PLL with neat diagram. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Explain the operation of PLL as frequency synthesizer. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Design a first order switched capacitive filter circuit. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Define antenna effect? How to avoid Antenna effect during layout. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) A 5 μ m/40nm MOSFET biased at 1mA exhibits a transconductance of 1/100 ohm . If the sheet resistance of the gate polysilicon is equal to 30ohm/square, what is the widest finger that the structure can incorporate while ensuring that the gate thermal noise voltage is 1/5th of the gate referred channel thermal noise voltage? (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_4445:CMOS mixed signals VLSI design
Course	

#	Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	17
3	3	CO 3	14
4	4	CO 4	6
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_4445:CMOS mixed signals VLSI design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)